AIRS-Assisted Wireless Power Transfer for Internet of Energy: Cybersecurity Challenges and Solutions

Vu Khanh Quy, Nguyen Minh Quy, Shaba Shaon, Md Raihan Uddin, and Dinh C. Nguyen

Abstract

Wireless power transfer (WPT) systems have emerged as a promising technology for the Internet of Energy to enable efficient energy distribution, especially the 6th-generation wireless networks (6G). To address this issue, aerial intelligent reflecting surfaces (AIRS) have been integrated with WPT, aiming to leverage their ability to dynamically manipulate wireless signals and improve energy transfer efficiency. Despite recent advancements in this domain, a comprehensive study addressing the cybersecurity challenges inherent in AIRS-powered WPT systems is missing. To bridge this crucial gap, our work is the first to systematically investigate four fundamental security threats in AIRS-powered WPT systems, including *eavesdropping and information leakage, spoofing and impersonation attacks, jamming and denial-of-service (DoS) attacks,* and *privacy concerns.* We analyze the vulnerabilities associated with these threats and propose effective countermeasures, laying the groundwork for the development of secure AIRS-powered WPT systems for the Internet of Energy in the 6G era.

Introduction

The connection for the adoption of a large-scale Internet of Things (IoT) presents a difficult task and challenges in providing global coverage over aerial, terrestrial, and maritime sectors. The advent of 6G networks is expected to provide all kinds of services and continuous network coverage for any kind of entity relying on breakout technologies. In 6G, one of the most challenging issues is power transfer for aerial and mobile IoT devices. From here, the Internet of Energy concept is formed, with the core component being WPT technology, which has emerged as a transformative technology, enabling the wireless transmission of energy to power devices without physical connections [1]. However, traditional WPT systems face challenges such as limited range, line-of-sight constraints, and energy inefficiency due to path loss and environmental obstacles. Recently, the intelligent reflecting surfaces (IRS) have emerged as a breakthrough development of 6G networks to fulfill their wide goals. The dynamic manipulation of radio waves, enabled by programmable metasurfaces, enhances transmission quality and addresses infrastructure limitations. Even in situations where terrestrial infrastructure faces challenges, IRSs placed in highly crowded urban areas can help users and base stations maintain LoS connections [2]. IRSs strategically positioned close to Autonomous Aerial Vehicles (AAVs) have many advantages, including lowered interference in crowded networks, targeted communication that maximizes energy consumption, and a notable increase in coverage range made possible by IRS beamforming capabilities. Using synergistic capabilities in mobility and environmental reconstruction to reduce congestion and fading, this dynamic team deftly addresses the issues presented by high-frequency bands, including THz and mmWave [3]. The amazing agility displayed by AAVs also helps to investigate creative ideas for IRS design and application. Full-angle reflection offers a remarkable degree of freedom in maximizing IRS configuration and performance. To address these limitations of WPT, *Aerial Intelligent Reflecting Surfaces (AIRS)* have been proposed as a groundbreaking solution [4]. AIRS combines IRSs and aerial platforms such as AAVs or high-altitude platforms. IRS is composed of programmable meta-materials capable of dynamically manipulating electromagnetic waves (e.g., adjusting phase, amplitude, and direction) to enhance signal propagation. When deployed on aerial platforms, IRS gains the added advantage of mobility and flexibility, enabling it to optimize energy transfer in dynamic and complex environments.

The core principle of AIRS involves reflecting and focusing electromagnetic waves from a power transmitter to intended receivers. The aerial platform positions the IRS to ensure optimal alignment, while the IRS dynamically adjusts the wave properties to maximize energy delivery efficiency. This approach not only extends the range of WPT systems but also overcomes traditional line-of-sight limitations, making it suitable for applications in remote or obstructed environments. AIRS-powered WPT provides the main advantages, as follows.

- **Extended Range:** AIRS enhances the range of WPT by focusing energy beams, allowing power to be transmitted efficiently over longer distances. This capability significantly

Digital Object Identifier: 10.1109/MNET.2025.3617988
Date of Current Version: 14 January 2026
Date of Publication: 23 October 2025

Vu Khanh Quy and Nguyen Minh Quy (corresponding author) are with the Faculty of Information Technology, Hung Yen University of Technology and Education, Hung Yen 170000, Vietnam; Shaba Shaon, Md Raihan Uddin, and Dinh C. Nguyen are with the Department of Electrical and Computer Engineering, The University of Alabama in Huntsville, Huntsville, AL 35899 USA.

reduces the impact of signal attenuation and increases the effective coverage area compared to traditional approaches.

- **Enhanced Efficiency:** By directing energy beams towards the receivers, AIRS minimizes energy loss and interference caused by environmental factors. This improves power transmission and reduces wasted energy.
- **Flexibility:** AIRS systems can dynamically adjust their position and orientation to adapt to shifting receiver locations or changes in environmental conditions. This enables the WPT system to operate stably and efficiently in highly dynamic or unpredictable environments.
- **Scalability:** The AIRS includes modules that allow for easy scaling. As demand increases or areas expand, the AIRS units can be added to ensure continuous and reliable power distribution.

AIRS-powered WPT is expected to play a key role in next-generation wireless networks, green energy solutions, and 6G communication systems, paving the way for intelligent, sustainable, and seamless power delivery. For example, AIRS-powered WPT can be used to wirelessly charge remote IoT sensor networks in smart agriculture, ensuring continuous operation without the need for battery replacements. Additionally, it can enable uninterrupted power delivery to autonomous drones and electric vehicles, enhancing mobility and operational efficiency in smart transportation systems.

However, AIRS-powered WPT also introduces unique security challenges due to its reliance on wireless communication, dynamic positioning, and open environments. Indeed, eavesdropping poses a significant threat in AIRS-powered WPT systems, as attackers can intercept transmitted signals to gather sensitive information about power transmission patterns or even device identities, potentially leading to unauthorized energy harvesting or data breaches. Additionally, jamming attacks can disrupt the efficiency of wireless power transfer by introducing malicious interference, which can degrade system performance, cause energy losses, or even render the entire WPT system unusable in critical applications.

Despite recent efforts, there is no existing work that thoroughly investigates the security challenges in AIRS-powered WPT systems. To fill this significant gap, this paper, for the first time, explores and analyzes potential security threats, vulnerabilities, and countermeasures in AIRS-powered WPT, providing a foundation for future 6G secure network designs. We focus on four popular and fundamental security challenges in AIRS-powered WPT systems, including eavesdropping and information leakage (the section "Eavesdropping and Information Leakage"), spoofing and impersonation attacks (the section "Spoofing and Impersonation Attacks"), jamming and DoS attacks (the section "Jamming and Denial-of-Service (DOS) Attacks"), and privacy concerns (the section "Privacy Concerns"). The overview of the AIRS-assisted WPT in 6G networks and its security scenarios, such as jamming and eavesdropping attacks are shown in Fig. 1.

Eavesdropping and Information Leakage

AIRS systems rely on wireless signals to transfer power and communicate with receivers. However, due to the open nature of wireless channels, these signals can be intercepted by malicious actors, leading to eavesdropping. Attackers can harvest energy intended for legitimate users or extract sensitive information, such as receiver locations or system configurations [5]. This section explores the key impacts of eavesdropping and information leakage in AIRS-powered WPT and discusses potential countermeasures to these threats.

Impact

- *Energy Harvesting Efficiency:* Through eavesdropping, attackers can passively intercept signals to steal energy and reduce the

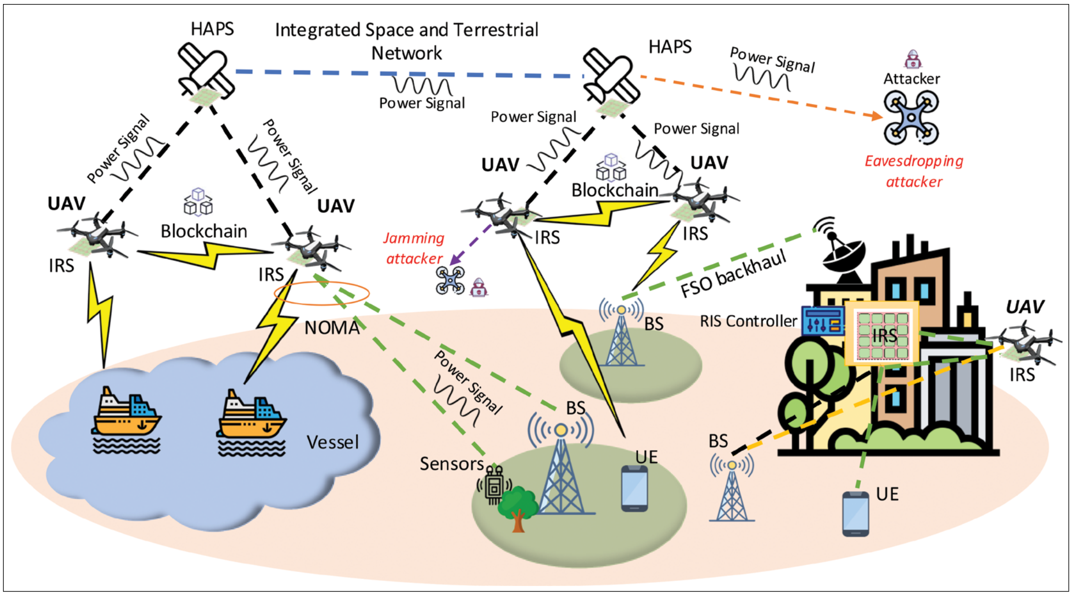

FIGURE 1. Overview of the AIRS-assisted WPT in 6G networks and its security scenarios, such as jamming and eavesdropping attacks.

efficiency of wireless power transmission. This leads to a reduction in energy availability for legitimate receivers and an increase in power consumption at the transmitter to compensate for losses. Furthermore, by intercepting IRS-reflected signals, attackers can add jamming signals to disrupt legitimate energy transfer, aiming to initialize spoofing attacks to mislead IRS configurations and reduce WPT efficiency.

- *Exploiting for Malicious Purposes:* In WPT, power signals often carry control information related to device authentication, energy allocation, and network protocols. The information leakage can lead to unauthorized access to control devices and communications protocols. All these allow attackers to change the power distribution plans and extract user behavior patterns, which could be exploited for malicious purposes.
- *Security Risks:* In the future, countless IoT systems and smart grids will rely on WPT. The eavesdropping and information leakage lead to further attacks. Moreover, exposing device identities leads to cyberattacks for purposes such as disrupting industrial and healthcare devices and causing system failures.
- *Steal WPT Signals:* In WPT, systems are made so attackers can obtain vital information undetectably by leveraging the WPT signals, therefore violating the integrity and privacy of the connection. Operating security and data confidentiality are seriously threatened by these assaults since they take advantage of weaknesses in WPT systems security mechanisms to gather sent energy and data.

The impacts highlight the risks of eavesdropping and information leakage in AIRS-assisted WPT. Besides the above issues, this also leads to higher operational costs and financial losses for businesses relying on WPT, and raises potential risks in critical domains such as smart healthcare and autonomous systems.

Solutions

- **Encryption:** In our vision, one of the most promising solutions for secure communication in AIRS-powered WPTs is quantum-based encryption methods, specifically Quantum Key Distribution (QKD). QKD leverages quantum mechanics principles to generate encryption keys unbreakable by any attack, including those from quantum computers. Any eavesdropping attempt disturbs the quantum state, enabling real-time detection of security breaches. By integrating QKD with post-quantum cryptographic algorithms, the system creates a hybrid security model that ensures resilience against both classical and quantum threats. This approach provides an extremely secure communication solution and resists spoofing attacks for AIRS-powered WPT.
- **Frequency Hopping:** This method rapidly switches transmission frequencies in a predefined pattern, making it extremely difficult for eavesdroppers to predict or jam the signal. By dynamically altering the operating frequency to reduce the risk of interference and malicious attacks. Furthermore, integrating AI-driven adaptive hopping further optimizes frequency selection based on real-time network conditions. This approach ensures secure, resilient, and efficient communication for AIRS-powered WPT in both energy transmission and data communication.
- **Beamforming Security:** This technique focuses energy precisely on authorized receivers. By dynamically adjusting the phase shifts of reflective elements, AIRS can steer energy beams toward intended users while minimizing leakage to unintended directions. This targeted energy transmission reduces the risk of eavesdropping and unauthorized power access, enhancing overall system security. Moreover, integrating advanced AI techniques optimized beamforming further improves efficiency by real-time adapting to environmental changes and receiver locations. This approach ensures secure, efficient, and limited interference for AIRS-powered WPT.
- **Authentication:** To ensure only authenticated devices can participate in the power transfer process, a robust authentication mechanism is used to verify devices before granting access to the system, such as public key infrastructure (PKI) or lightweight authentication schemes. AIRS can further enhance security by dynamically adjusting power distribution based on real-time authentication status, preventing unauthorized access. AI-driven anomaly detection can identify and block fraudulent devices attempting to bypass authentication. This approach guarantees a secure and efficient AIRS-powered WPT.

In summary, addressing eavesdropping and information leakage in AIRS-powered WPT systems requires a layered defense strategy. By combining these solutions, AIRS-assisted WPT can achieve a more secure, efficient, and trustworthy foundation for 6G IoE deployment.

Spoofing and Impersonation Attacks

In AIRS-powered WPT systems, spoofing and impersonation attacks pose significant security threats and strictly impact system performance, energy efficiency, and data integrity. For instance, attackers can spoof legitimate receivers by mimicking their signals or credentials, tricking the AIRS into directing energy to unauthorized locations [6]. This can be achieved by replaying legitimate signals or exploiting weak authentication mechanisms. In this section, we discuss impacts and countermeasures.

Impact

- *Energy Harvesting Manipulation:* Attackers can impersonate legitimate receivers and divert the transmitted power to unauthorized devices or malicious actors. This reduces the power available to intended users, leading to energy starvation or even a denial of service (DoS) for legitimate energy receivers.
- *Interference and Jamming:* Spoofing attacks can inject channel state misinformation, causing the IRS to misconfigure its reflection

coefficients. This results in beam misalignment, lowering energy transfer efficiency, and wireless communications. Moreover, attackers impersonate an authorized node to inject malicious signals into the system. This interferes with energy transfer and data transmission, reducing the system's effectiveness.

- *System Control Hijacking:* If an attacker gains access to the IRS control system via impersonation, they can manipulate the phase shifts to redirect energy or disrupt legitimate users. This could lead to unintended energy distribution or even cause system instability.

Spoofing and impersonation attacks can severely undermine both energy efficiency and system integrity. Collectively, these impacts not only reduce performance and reliability but also pose significant risks to the security of critical infrastructures that depend on AIRS-powered WPT.

Solutions

- **Strong Authentication:** Integrating multi-factor authentication (MFA) and cryptographic protocols is a possible countermeasure against spoofing and impersonation attacks in AIRS-powered WPT systems. MFA requires multiple verification factors, such as biometrics and time-based one-time passwords, making it difficult for attackers to impersonate legitimate users. Cryptographic protocols, including digital signatures, ensure data authenticity and integrity by verifying the identities of communicating entities. These measures prevent malicious actors from injecting false credentials or mimicking legitimate devices to gain unauthorized access. By combining MFA and digital signatures, the system establishes a robust security framework that effectively mitigates spoofing and impersonation threats.
- **Challenge-Response Mechanisms:** The approach provides an effective countermeasure against spoofing and impersonation attacks in AIRS-powered WPT by verifying receiver identity. In this approach, the system generates a unique challenge that only a legitimate receiver can correctly respond to using a pre-shared cryptographic key. Attackers attempting to impersonate the receiver will fail to generate the correct response, preventing unauthorized access. By integrating cryptographic challenge-response protocols, such as hash-based message authentication codes or public-key cryptography, the system ensures secure authentication. This dynamic verification process strengthens WPT security, making it resistant to identity-based attacks.
- **Location Verification:** The solution strengthens security in AIRS-powered WPT by using GPS or other localization techniques to counter spoofing and impersonation attacks. By verifying the receiver's physical location, the system ensures that only authorized devices in predefined areas can access wireless power. Attackers attempting to spoof identities from unauthorized locations will be detected and blocked from the network. Advanced localization methods, such as time-of-flight measurements or angle-of-arrival techniques, enhance accuracy and prevent location spoofing. This approach adds an extra security layer, ensuring that only legitimate receivers in approved locations can participate in the WPT process.
- **Dynamic Credentials:** The method provides a strong defense against spoofing and impersonation attacks in AIRS-powered WPT by using time-based or session-based authentication. These credentials change frequently, ensuring that even if an attacker intercepts authentication data, it becomes useless after a short period. The TOTP and session keys prevent replay attacks by ensuring that credentials expire after each use. By integrating cryptographic techniques, such as hash functions or asymmetric encryption, dynamic credentials enhance security and prevent unauthorized access. This approach ensures that only legitimate receivers with valid, up-to-date credentials can authenticate and participate in the WPT process.

Mitigating spoofing and impersonation in AIRS-powered WPT systems requires a multi-directional security approach. Together, these solutions create a robust defense framework, preventing unauthorized devices from diverting power or manipulating system control.

Jamming and Denial-of-Service (DoS) Attacks

Jamming and DoS attacks pose serious threats to AIRS-powered WPT systems. While jamming interferes with beamforming [7], DoS attacks overload the system, leading to disruption of systems and communication control [8]. In this section, we present impacts and countermeasures against jamming and DoS attacks on AIRS-powered WPT.

Impact

- *Disruption of Power Transfer:* Interference from jamming attacks can block or degrade power transmission efficiency, leading to energy shortages for legitimate receivers and even disrupting power transfer. Moreover, attacks can lower the effectiveness of the AIRS in optimizing wireless power distribution and reducing overall system efficiency.
- *System Downtime:* DoS attacks targeting control signals can result in system communication and control loss, preventing the system from dynamically adjusting power distribution and uncontrolled or inefficient energy transfer. Moreover, repeated attacks can cause delays in authentication, access control, and power delivery, leading to intermittent service disruptions or system downtime.
- *Damaged Devices and Unauthorized Access:* Attackers may exploit system weaknesses during a DoS attack to bypass authentication mechanisms, potentially leading to unauthorized energy consumption, unauthorized access to data, or damage to the devices.
- *Intercept WPT signals:* Attackers may seek to purposefully disrupt energy transmission by generating noise or signals at like frequencies. These attacks seriously impair the receiver's capacity to properly decode

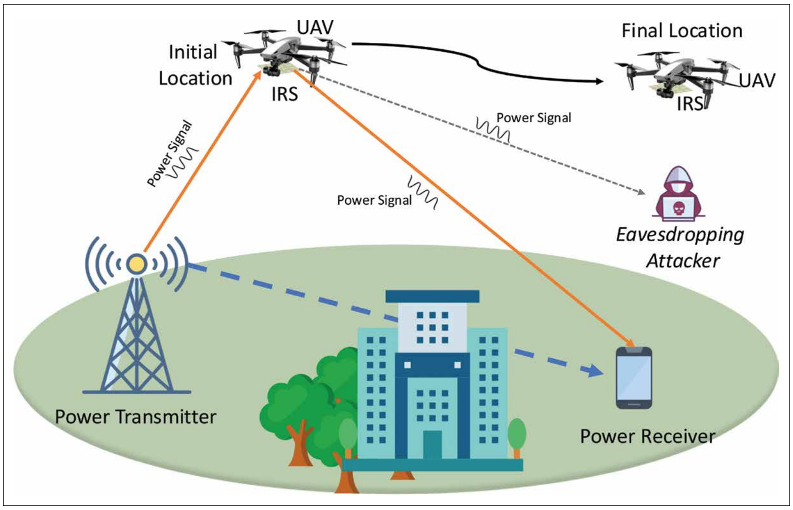

FIGURE 2. AAV-assisted WPT systems under the presence of eavesdropping attacks.

and make use of the sent power, therefore degrading system performance, as shown in Fig. 2

The attacks present critical threats to the reliability and stability of AIRS-powered WPT. They have serious impacts on both system availability and trustworthiness. This also shows the need for timely and robust countermeasures.

Solutions

- **Frequency Diversity:** The method enhances the resilience of AIRS-powered WPT systems against jamming and DoS attacks relying on utilizing multiple frequency bands or spread-spectrum techniques. By dynamically switching between different frequencies, the system can avoid interference and maintain continuous power transmission even in the presence of jamming signals. Spread-spectrum techniques, such as frequency hopping and direct sequence spread spectrum (DSSS), distribute the signal over a wider bandwidth, making it harder for attackers to jam the entire communication channel [9]. This approach ensures reliable energy transfer while minimizing the risk of service disruption due to malicious interference. Implementing frequency diversity strengthens the security and robustness of WPT systems, preventing attackers from compromising power delivery.
- **Jamming Detection:** The method enhances the security of AIRS-powered WPT systems by deploying real-time monitoring mechanisms to identify and counter-jamming attacks. By continuously analyzing signal patterns and power transmission efficiency, the system can detect anomalies caused by interference or malicious disruptions. Advanced ML-based anomaly detection techniques and signal-to-noise ratio (SNR) monitoring enable rapid identification of jamming attempts. Once a jamming attack is detected, countermeasures such as frequency switching, adaptive beamforming, or alerting security systems can be triggered to mitigate its impact [10]. This proactive approach ensures uninterrupted power delivery and strengthens the resilience of AIRS-powered WPT systems against DoS attacks.
- **Adaptive Beamforming:** This technique allows AIRS-powered WPT systems against jamming and DoS attacks, relying on dynamically adjusting beamforming patterns to avoid interference. By steering energy beams toward legitimate receivers while nullifying jamming sources, the system ensures efficient and secure power delivery. Real-time signal processing techniques enable the system to detect interference and reconfigure phase shifts in IRSs to minimize jamming impact [11]. This approach enhances power transfer reliability, even in hostile environments with active attackers. By continuously optimizing beamforming patterns, the system maintains robust performance and resists malicious signal disruptions.
- **Redundancy:** The solution enhances the resilience of AIRS-powered WPT against jamming and DoS attacks by deploying multiple active IRS platforms. If this one platform is compromised due to interference, alternative platforms can take over to maintain uninterrupted power delivery. This distributed approach reduces the system's dependency on a single reflective surface, ensuring that power transfer remains stable even in the presence of malicious disruptions. By intelligently coordinating multiple active IRS units, the system can dynamically reroute energy beams to bypass jamming sources. This method improves fault tolerance, enhances reliability, and strengthens the overall security of AIRS-powered WPT.

Countering jamming and DoS attacks in AIRS-powered WPT is a challenge and requires resilient and adaptive defense mechanisms. These solutions establish a multi-layered defense that safeguards both power efficiency and service availability.

Privacy Concerns

Privacy concerns in AIRS-powered WPT systems can lead to serious security and trust issues. AIRS systems may collect and process sensitive data, such as the location and usage patterns of receivers, to optimize power transfer. If this data is not properly secured, it could be exploited to track users or infer sensitive information [12]. In this section, we highlight privacy concerns and countermeasures.

Impact

- *Violation of User Privacy:* AIRS systems rely on real-time environmental sensing and data collection to optimize wireless power transfer. If this data is intercepted, attackers could track user movements, behaviors, or device usage patterns. Moreover, unauthorized access to location-based or energy consumption data could expose personal and sensitive information.
- *Potential Misuse of Data:* Attackers or malicious entities could exploit collected data for control, targeted cyberattacks, or unauthorized monitoring. If AIRS systems are

integrated into smart homes/cities or IIoT systems, leaked data could be used for identity theft or unauthorized profiling of users.
- *Loss of Trust in the System:* End-users and organizations may lose confidence in AIRS-powered WPT if they perceive them as vulnerable to privacy breaches. A lack of transparent data handling policies can result in challenges in adopting the solution to pervasive applications such as smart healthcare, smart cities, and financial transactions.

Privacy concerns in AIRS-powered WPT extend beyond technical risks to affect user trust and pervasive system adoption and create both security vulnerabilities and social barriers, making strong safeguards essential for large-scale deployment.

Solutions

- **Data Anonymization:** This is a crucial method to protect privacy in AIRS-powered WPT systems by removing personally identifiable information (PII) from collected data. This process ensures that sensitive user details, such as location, device identity, and energy consumption patterns, cannot be linked to specific individuals. Data masking, generalization, and differential privacy help obscure or randomize identifiable attributes while retaining useful system insights. By anonymizing data, systems can minimize privacy risks while still optimizing wireless power transfer efficiency [13]. Implementing robust anonymization methods enhances user trust and ensures compliance with data protection regulations.
- **Access Control:** This is one of the most essential approaches for protecting sensitive data by restricting unauthorized access. Blockchain has emerged as a promising solution for access control, offering a decentralized and tamper-resistant framework that enhances security and transparency [14]. By utilizing smart contracts, access policies can be dynamically enforced, ensuring automated verification and revocation without manual intervention. Additionally, blockchain's immutability ensures that all access logs remain secure and auditable, preventing unauthorized alterations and improving trust in system security. The combination of blockchain's decentralized nature, transparency, and automated enforcement makes it a superior alternative to conventional methods, strengthening the resilience of AIRS-powered WPT systems against data breaches and unauthorized access attempts.
- **Encryption:** The procedure is essential for ensuring that sensitive data remains protected from unauthorized access. Data must be encrypted in both states, in transit and at rest. Data encryption in transit safeguards wireless communications between devices, preventing interception or eavesdropping by malicious actors. In contrast, Encryption at rest ensures that stored data, including user credentials and system logs, remains secure even if physical storage is compromised. In our vision, implementing robust encryption protocols, such as AES-256 for rest data and TLS for data in transit, strengthens privacy in AIRS-powered WPT networks.
- **Privacy-Preserving Algorithms:** Federated learning (FL) and several differential privacy reduce the need for centralized data collection. Consequently, they help mitigate privacy concerns. Indeed, FL enables decentralized model training, ensuring that sensitive user data remains on local devices rather than being transmitted to a central server. In contrast, differential privacy adds controlled noise to datasets, preventing the identification of individual users while still allowing accurate system analysis and optimization. By implementing these techniques, AIRS-powered WPT systems can enhance privacy and security while maintaining high system efficiency.
- **User Consent:** This is a fundamental privacy safeguard mechanism in AIRS-powered WPT systems, ensuring transparency in data collection practices. Informing users about what data is collected, how it is used, and who can access it helps build trust and encourages responsible data management. Implementing clear and accessible consent mechanisms, such as opt-in agreements and granular privacy settings, allows users to control their personal information. Furthermore, regularly updating privacy policies and allowing users to modify or revoke their consent enhances compliance with data protection regulations [15]. By prioritizing user awareness and informed consent, systems can mitigate privacy concerns.

Addressing privacy concerns requires robust data protection and user-centric safeguards through mechanisms such as data anonymization, access control, and end-to-end encryption. Besides, federated learning reduces centralized data while maintaining system optimization. These solutions strengthen trust, compliance, and adoption, paving the way for driving the AIRS-powered WPT into critical infrastructures.

Attack Method	Impacts	Countermeasures
Eavesdropping & Information Leakage	· Energy harvesting by attackers, reducing efficiency · Information leakage of control signals and user behavior · Unauthorized access and system disruption	· Quantum-based encryption · Frequency hopping & Beamforming security · Robust authentication (PKI, anomaly detection)
Spoofing & Impersonation Attacks	· Divert power to unauthorized devices · Interference & beam misalignment · Control hijacking of IRS configuration	· Strong authentication (MFA, digital signatures) · Challenge-response mechanisms & Location verification · Dynamic credentials (time/session-based)
Jamming Attacks & DoS Attacks	· Disruption of power transfer · System downtime & Intercept WPT signals · Damaged devices and unauthorized access	· Frequency diversity & Jamming detection · Adaptive beamforming & Redundancy
Privacy Concerns	· Violation of user privacy via data collection · Misuse of location/energy usage data · Loss of user/system trust	· Data anonymization & Access Control · Strong encryption & Privacy-Preserving · User Consent

TABLE 1. Main Security Attacks, Impacts, and Countermeasures in AIRS-powered WPT Systems.

Conclusion

In this paper, we investigate cybersecurity issues for the Internet of Energy, especially in AIRS-powered WPT systems. We highlighted four main security threats: *eavesdropping and information leakage, spoofing and impersonation attacks, jamming and DoS attacks*, and *privacy concerns*. Moreover, we also identify key vulnerabilities associated with these threats and propose effective mitigation strategies, contributing to the foundation of secure AIRS-powered WPT systems in the 6G era. We believe that our work sheds light on new insights into the unique security challenges posed by AIRS-powered WPT and will encourage further exploration into effective defense strategies. By addressing these security concerns, we aim to pave the way for the safe and reliable deployment of AIRS-assisted wireless power transfer for the Internet of Energy era.

References

[1] M. M. Wang, J. Zhang, and X. You, "Ultra-low-energy Internet of Things from a network and device perspective," *IEEE Commun. Mag.*, vol. 60, no. 12, pp. 50–55, Dec. 2022.

[2] Y. Cheng, W. Peng, and T. Jiang, "Self-sustainable RIS aided wireless power transfer scheme," *IEEE Trans. Veh. Technol.*, vol. 72, no. 1, pp. 881–892, Jan. 2023.

[3] X. Pang et al., "When UAV meets IRS: Expanding air-ground networks via passive reflection," *IEEE Wireless Commun.*, vol. 28, no. 5, pp. 164–170, Oct. 2021.

[4] H. Ren et al., "Energy minimization in RIS-assisted UAV-enabled wireless power transfer systems," *IEEE Internet Things J.*, vol. 10, no. 7, pp. 5794–5809, Apr. 2023.

[5] F. Tao and D. Ye, "Active eavesdropping attack scheduling for cyber–physical systems with operation constraints," *IEEE Internet Things J.*, vol. 12, no. 3, pp. 3067–3075, Feb. 2025.

[6] B. Zheng et al., "Intelligent reflecting surface-enabled anti-detection for secure sensing and communications," *IEEE Wireless Commun.*, vol. 32, no. 2, pp. 156–163, Apr. 2025.

[7] D. Adhikary, J. Plusquellic, and E. E. Tsiropoulou, "Jamming attacks detection and ejection in over the air computation concentrated solar power systems," *IEEE Internet Things J.*, early access, Sep. 4, 2025, doi: 10.1109/JIOT.2025.3606344.

[8] T. Wu, J. Liu, L. Xue, and Y. Wu, "Fixed-time synchronization of multilayer complex networks under denial-of-service attacks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 9, pp. 3519–3523, Sep. 2023.

[9] B. M. Todorović, N. M. Stojanović, and G. S. Velikić, "Is direct sequence spread spectrum modulation promising for AI-based design of 6G networks?" *IEEE Consum. Electron. Mag.*, vol. 13, no. 5, pp. 7–10, Sep. 2024.

[10] A. Meftah et al., "Federated learning-enabled jamming detection for stochastic terrestrial and non-terrestrial networks," *IEEE Trans. Green Commun. Netw.*, vol. 9, no. 1, pp. 271–290, Mar. 2025.

[11] S. Li and Q. Song, "User-adaptive IRS beamforming design in hybrid wireless powered communication networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 7, pp. 10755–10760, Jul. 2024.

[12] Q. Pan et al., "Differential privacy and IRS empowered intelligent energy harvesting for 6G Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 22, pp. 22109–22122, Nov. 2022.

[13] M. Milani, Y. Huang, and F. Chiang, "Data anonymization with diversity constraints," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3603–3618, Apr. 2023.

[14] D. C. Nguyen et al., "Cooperative task offloading and block mining in blockchain-based edge computing with multi-agent deep reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 22, no. 4, pp. 2021–2037, Apr. 2023.

[15] A. F. Eliyan et al., "Informed consent in IoT environments: Challenges and a futuristic incentives-based model," *IEEE Internet Things Mag.*, vol. 7, no. 1, pp. 112–119, Jan. 2024.

Biographies

Vu Khanh Quy was born in Hai Duong, Vietnam, in 1982. He received the M.Sc. and Ph.D. degrees from the Posts and Telecommunications Institute of Technology in 2012 and 2021, respectively. He is currently a Senior Lecturer at the Hung Yen University of Technology and Education, Vietnam. He is the first author of over 30 published papers in reputed journals and conferences. His research interests include wireless communication, mobile ad-hoc networks, high-performance computing, edge computing, the Internet of Things, next-generation networks, the IoT ecosystems, and communication networks.

Nguyen Minh Quy (minhquy@utehy.edu.vn) received the B.S. degree in information technology from the Hanoi University of Science and Technology, the master's degree in software engineering from the VNU University of Engineering and Technology, Vietnam, and the Ph.D. degree in software engineering from the Hanoi University of Science and Technology in 2015. He is the Vice-Rector of the Hung Yen University of Technology and Education. His research interests include high-performance computing and mobile communication networks.

Shaba Shaon is currently pursuing the Ph.D. degree with the College of Engineering, The University of Alabama in Huntsville, Huntsville, AL, USA. She is with the Networking, Intelligence, and Security Research Laboratory, The University of Alabama in Huntsville. Her research has been submitted to renowned IEEE journals and conferences, including IEEE Transactions on Vehicular Technology, IEEE Transactions on Wireless Communications, and the IEEE CCNC Conference. Her research interests include federated learning, multimodal learning, wireless communication, networking, and optimization algorithms.

Md Raihan Uddin (Graduate Student Member, IEEE) received the bachelor's degree in electronics and telecommunication engineering from Daffodil International University, Dhaka, Bangladesh. He is currently pursuing the master's degree in computer engineering with The University of Alabama in Huntsville, USA. He is a Researcher at the Networking, Intelligence, and Security Laboratory, The University of Alabama in Huntsville, under the guidance of Dr. Dinh C. Nguyen. He has developed a robust foundation in multiple aspects of technology, particularly in machine learning, artificial intelligence, privacy and security, and wireless communication.

Dinh C. Nguyen received the Ph.D. degree in computer science from Deakin University, Australia, in 2021. He worked as a Post-Doctoral Research Associate at Purdue University, USA, from 2022 to 2023. He is an Assistant Professor at the Department of Electrical and Computer Engineering, The University of Alabama in Huntsville, USA. He has published over 50 papers (including over 25 first-authored papers) on top-tier IEEE/ACM conferences and journals. His current research interests include federated machine learning, the Internet of Things, wireless networking, and security. He is the Technical Program Chair of the 6G Connectivity Workshop at the IEEE Wireless Communications and Networking Conference (WCNC) in 2024. He received the Best Editor Award from the IEEE Open Journal of Communications Society in 2023. He is an Associate Editor of IEEE Internet of Things Journal and IEEE Open Journal of the Communications Society and a Lead Guest Editor for the Special Issue on Federated Learning for Industrial Internet of Things of *IEEE Internet of Things Magazine*.